

Platform Administration & Billing Module Plan

Companion Document to SaaS High School Management System v4

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Super Admin Dashboard • Subscription Lifecycle • Payment Integration

Pricing Tiers • Currency & Conversion • Notification System

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1. Module Overview & Purpose

This module is the central nervous system of the SaaS platform. It gives the platform owner (Super Admin) complete control over every school subscription, payment, pricing, promotion, and system-wide configuration from a single unified dashboard.

1.1 Core Objectives

- Single dashboard to manage ALL platform operations
- Full billing lifecycle: trial → active → suspended → archived
- Multiple payment methods: Visa card, MTN Mobile Money, Orange Money
- Dual currency support: USD and LRD with real-time conversion
- Automated notifications for subscription expiry and deactivation warnings
- Complete pricing control: create, edit, discount, and promote plans
- Scalable admin roles for future team growth

1.2 Design Philosophy

Start lean. One Super Admin (the platform owner) manages everything. As the platform grows, the same dashboard supports adding staff with role-based permissions. No separate tools needed — everything lives in one place.

2. Super Admin Dashboard

The Super Admin Dashboard is the single control center for the entire SaaS platform. It provides a bird's-eye view of all operations with drill-down capability into any area.

2.1 Dashboard Overview Panel

Widget	Description	Data Source
Total Schools	Count of all registered schools (active + inactive)	schools table
Active Subscriptions	Schools with current paid plans	subscriptions table
Revenue This Month	Total payments received (USD + LRD converted)	payments table
Expiring Soon	Schools whose subscription ends within 7 days	subscriptions table
Overdue Payments	Schools past due date with no payment	invoices table
New Signups (30d)	Schools registered in the last 30 days	schools table
System Health	Server uptime, storage usage, API response times	monitoring

2.2 Dashboard Sections

School Management

- View all schools with status badges (Active, Trial, Suspended, Archived)
- Search/filter by name, location, plan, status
- Click any school to see full details: subscription, payments, users, usage
- Manually activate/deactivate any school
- Edit school profile, domain, branding settings

Billing & Payments

- View all invoices across all schools
- Payment history with method breakdown (Visa/MTN/Orange)
- Manual payment recording for bank transfers
- Refund processing
- Revenue reports by period, plan, payment method

Pricing & Plans

- Create/edit/delete pricing plans
- Set prices in USD (LRD auto-calculated)
- Manage discounts and promotional offers
- Set trial period duration
- Configure grace periods for late payments

Notifications Center

- View all sent notifications
- Configure notification templates and timing
- Manual broadcast to all schools or filtered groups
- Email delivery status tracking

System Settings

- Platform branding (logo, colors, name)
- Default currency and conversion rate source
- Email sender configuration
- Domain management
- Backup and maintenance controls

2.3 Super Admin Capabilities

The Super Admin has unrestricted access to every function:

Capability	Description
Create/Edit/Delete Plans	Full pricing control including custom plans for specific schools
Apply Discounts	Percentage or fixed amount, time-limited or permanent
Override Subscriptions	Manually extend, upgrade, downgrade any school
Process Payments	Record manual payments, issue refunds
Deactivate Schools	Instant suspension with data preservation
Edit All Content	Modify any notification template, email, or system text
View All Data	Access any school's data for support purposes
Manage Domains	Assign/change subdomains and custom domains
Export Reports	Download CSV/PDF reports of any data
Create Admin Users	Add staff accounts when ready to scale (future)

3. Pricing Tiers & Plans

All prices are set in USD as the base currency. LRD equivalent is calculated automatically using real-time exchange rates. The Super Admin can edit all pricing at any time.

3.1 Plan Structure

Plan	Billing	Price (USD)	Student Limit	Features
Basic Monthly	Monthly	\$25/mo	Up to 200	Core modules only
Basic Yearly	Yearly	\$250/yr (save 17%)	Up to 200	Core modules only
Standard Monthly	Monthly	\$50/mo	Up to 500	Core + Reports + Letters
Standard Yearly	Yearly	\$500/yr (save 17%)	Up to 500	Core + Reports + Letters
Premium Monthly	Monthly	\$100/mo	Up to 1500	All modules + Priority support
Premium Yearly	Yearly	\$1,000/yr (save 17%)	Up to 1500	All modules + Priority support
Premier (Enterprise)	Custom	Contact Us	Unlimited	Lifetime partnership + Dedicated support

3.2 Premier Partnership Plan

The Premier plan is for schools that want a lifetime partnership with the platform. This is not a self-service plan — schools contact the Super Admin directly to negotiate terms.

- Custom pricing based on school size and needs
- One-time setup fee + reduced ongoing rate, OR lump-sum lifetime fee
- Dedicated account manager (when team grows)
- Custom feature development priority
- White-label branding included
- SLA with guaranteed uptime

3.3 Editable Plan Properties

Every aspect of a plan can be edited by the Super Admin at any time:

Property	Editable?	Notes
Plan Name	Yes	Rename plans freely
Price (USD)	Yes	Changes apply to new subscriptions only
Price (LRD)	Auto-calculated	Based on current exchange rate
Student Limit	Yes	Can increase/decrease per plan

Feature Set	Yes	Toggle modules on/off per plan
Trial Duration	Yes	Default 30 days, configurable 7–90 days
Grace Period	Yes	Default 7 days after expiry
Billing Cycle	Yes	Monthly, Yearly, or Custom
Visibility	Yes	Show/hide plans from public pricing page
Custom Plans	Yes	Create one-off plans for specific schools

4. Payment Methods & Integration

The platform supports three payment methods to accommodate the Liberian market:

4.1 Visa/Mastercard (Stripe)

- Integration via Stripe payment gateway
- Supports Visa, Mastercard, and international cards
- Automatic recurring billing for monthly/yearly plans
- PCI-DSS compliant — card data never touches our servers
- Instant payment confirmation
- Automatic retry for failed payments (3 attempts over 7 days)

4.2 MTN Mobile Money

- Integration via MTN MoMo API (Liberia)
- School admin enters phone number, receives payment prompt on phone
- Payment confirmed via API callback
- Supports both USD and LRD wallets
- Manual reconciliation dashboard for failed callbacks
- Transaction reference stored for audit

4.3 Orange Money

- Integration via Orange Money API (Liberia)
- Same flow as MTN: enter number → approve on phone → confirmed
- Supports LRD primarily
- Webhook-based confirmation
- Fallback: manual payment recording by Super Admin

4.4 Payment Processing Flow

1. School selects plan and billing cycle
2. School chooses payment method (Visa / MTN / Orange)
3. Payment is processed through respective gateway
4. On success: invoice generated, subscription activated
5. On failure: school notified, retry scheduled (card) or manual retry (mobile money)
6. Super Admin can see all transactions in real-time

4.5 Manual Payment Recording

For bank transfers or cash payments, the Super Admin can manually record payments:

- Select school → Record Payment → Enter amount, method, reference
- System auto-generates invoice and activates/extends subscription

- Audit trail shows “Manually recorded by Super Admin”

5. Currency Management (USD/LRD)

5.1 Dual Currency Display

All prices throughout the platform display in both USD and LRD:

- Base price always set in USD by Super Admin
- LRD equivalent calculated automatically using current exchange rate
- Example: \$25 USD = L\$4,875 LRD (at rate of 1 USD = 195 LRD)

5.2 Exchange Rate Management

Method	Description	Priority
Automatic API	Fetch from exchangerate-api.com or similar every 6 hours	Primary
Manual Override	Super Admin sets a fixed rate manually	Override
Fallback Rate	Last known rate used if API fails	Fallback

5.3 Rate Configuration (Super Admin)

- View current rate with last update timestamp
- Toggle between automatic and manual rate
- Set rate update frequency (1hr, 6hr, 12hr, 24hr)
- Rate change history log
- Alert if rate changes more than 10% in 24 hours (unusual volatility)

5.4 Payment Currency Rules

- Visa/Mastercard: charged in USD
- MTN Mobile Money: charged in USD or LRD (based on wallet type)
- Orange Money: charged in LRD (converted from USD at current rate)
- All revenue reports can toggle between USD and LRD views

6. Subscription Lifecycle

6.1 Lifecycle States

State	Description	School Access	Duration
Trial	Free trial period after signup	Full access	30 days (configurable)
Active	Paid subscription current	Full access	Until billing period ends
Grace Period	Subscription expired, payment pending	Limited access (read-only)	7 days (configurable)
Suspended	No payment after grace period	Login blocked, data safe	Until payment or 90 days
Archived	Long-term non-payment	No access, data in cold storage	Recoverable for a fee
Cancelled	School voluntarily cancelled	Access until period ends	Until billing period ends
Premier	Lifetime partnership	Full access forever	Indefinite

6.2 Automatic State Transitions

The system automatically transitions schools between states:

Trial (30 days) → If payment: Active | If no payment: Grace Period (7 days) → If payment: Active | If no payment: Suspended → After 90 days: Archived

Every transition triggers a notification to the school and logs in the audit trail.

6.3 Super Admin Overrides

- Extend trial for any school (e.g., extra 15 days for a promising lead)
- Skip grace period and suspend immediately
- Reactivate suspended school instantly after payment
- Move school to Premier status
- Override any automatic transition

7. Notification & Alerting System

Automated notifications keep schools informed about their subscription status and prevent unexpected deactivations.

7.1 Notification Schedule

Trigger	When	Channel	Message Summary
Trial Ending	7 days before trial expires	Email	Your free trial ends in 7 days. Subscribe to continue.
Trial Ending	3 days before trial expires	Email + SMS	3 days left! Choose a plan now.
Trial Expired	Day of expiry	Email + SMS	Trial ended. Subscribe or lose access in 7 days.
Payment Due	7 days before billing date	Email	Your subscription renews on [date]. Ensure payment is ready.
Payment Due	3 days before billing date	Email	Payment due in 3 days for [plan name].
Payment Due	Day of billing	Email + SMS	Payment due today. Process now to avoid interruption.
Payment Failed	Immediately	Email + SMS	Payment failed. Please update payment method.
Payment Retry	Day 3 and Day 5	Email	We'll retry your payment. Update method if needed.
Grace Period Start	Day after expiry	Email + SMS	Your subscription has expired. You have 7 days to pay.
Grace Period Warning	Day 5 of grace	Email + SMS	2 days left! Your school site will be deactivated.
Suspension Notice	Day of suspension	Email + SMS	Your school site has been deactivated. Pay to reactivate.
Payment Received	Immediately	Email	Payment confirmed! Your subscription is active until [date].
Plan Upgrade	Immediately	Email	You've been upgraded to [plan]. New features unlocked!
Price Change	30 days before effective	Email	Your plan price will change on [date]. See details.

7.2 Notification Templates (Editable)

All notification content is editable by the Super Admin:

- Email subject line and body (rich text editor)
- SMS message (160 character limit)

- Dynamic variables: {{school_name}}, {{plan_name}}, {{expiry_date}}, {{amount_usd}}, {{amount_lrd}}, {{payment_link}}
- Preview before saving
- Test send to Super Admin email

7.3 Manual Notifications

- Broadcast to all schools (e.g., maintenance notice, new feature)
- Send to filtered group (e.g., all Basic plan schools)
- Individual school notification
- Notification delivery log with read receipts

8. Discount & Promotion Management

The Super Admin has full control over discounts and promotional pricing:

8.1 Discount Types

Type	Description	Example
Percentage Discount	Reduce price by X%	20% off first 3 months
Fixed Amount	Reduce price by \$X	\$10 off monthly plan
Coupon Code	Redeemable code for discount	Code: LAUNCH2026 for 15% off
School-Specific	Custom price for one school	Lincoln High gets \$40/mo instead of \$50
Bulk Discount	Discount for multiple school signups	Register 3+ schools, get 25% off each
Seasonal Promotion	Time-limited campaign	Back-to-school: 30% off yearly plans in August
Referral Discount	Discount when school refers another	Refer a school, both get \$5 off next month
Loyalty Discount	Automatic after X months	After 12 months, get 10% off renewal

8.2 Promotion Configuration

Each promotion has these editable properties:

- Name and description
- Discount type and value
- Start date and end date
- Applicable plans (all or specific)
- Usage limit (total redemptions or per-school)
- Coupon code (auto-generated or custom)
- Stackable: can this combine with other discounts?
- Active/Inactive toggle

8.3 Promotion Dashboard

- List all promotions with status (Active, Scheduled, Expired)
- Usage stats: how many times redeemed, revenue impact
- Quick duplicate: copy an existing promotion and modify
- Promotion performance chart: signups attributed to each promo

9. School Onboarding Flow

When a new school signs up, the system follows this automated flow:

9.1 Self-Service Signup

Step 1: School visits the platform website

Step 2: Fills registration form (school name, location, admin email, phone)

Step 3: Email verification sent

Step 4: School admin verifies email and sets password

Step 5: System auto-creates:

- School profile with unique school_id
- Subdomain: schoolname.yoursystem.com
- 30-day free trial subscription
- Default letter templates (23 pre-loaded)
- Admin account for the school

Step 6: School admin logs in and starts configuring their school

Step 7: Super Admin receives notification of new signup

9.2 Domain Setup

- Every school gets: schoolname.yoursystem.com (automatic)
- Optional: school connects their own domain (e.g., www.lincolnhigh.edu.lr)
- Domain connection requires DNS CNAME record pointing to your platform
- SSL certificate auto-provisioned via Let's Encrypt
- Super Admin can manage all domains from the dashboard

9.3 White-Label Branding

Each school's site is white-labeled:

- School's logo, colors, and name displayed
- No mention of the SaaS platform name visible to students/parents
- Custom login page with school branding
- Under the hood, it's your system — just like Lovable apps

10. Admin User Management (Future Scaling)

Initially, one Super Admin (the platform owner) manages everything. As the platform grows, the dashboard supports adding team members:

10.1 Planned Admin Roles

Role	Access Level	When to Hire
Super Admin (You)	Full unrestricted access to everything	Day 1 — this is you
Billing Manager	Payments, invoices, subscriptions, refunds	50+ schools
Support Agent	View school data, respond to tickets, basic troubleshooting	100+ schools
Technical Admin	System health, backups, server management	200+ schools
Sales/Onboarding	School registration, trial management, promotions	When actively marketing

10.2 Role-Based Access Control

When you add staff, each role has defined permissions:

- Billing Manager: cannot change system settings or delete schools
- Support Agent: read-only access to school data, can escalate issues
- Technical Admin: server access but no billing or school data editing
- All actions logged with user ID for accountability
- Super Admin can revoke any role instantly

10.3 For Now: Single Admin Dashboard

Until the platform grows, the Super Admin dashboard gives you everything in one place. No need to switch between tools. The role-based system is built in but dormant — flip it on when you're ready to hire.

11. Database Schema

Core tables for the billing and administration module:

subscription_plans

Column	Type / Constraint
id	UUID, PK
name	VARCHAR(100)
slug	VARCHAR(50), unique
description	TEXT
price_usd	DECIMAL(10,2)
billing_cycle	ENUM(monthly, yearly, custom, lifetime)
student_limit	INTEGER
features	JSONB
is_active	BOOLEAN
is_visible	BOOLEAN
trial_days	INTEGER DEFAULT 30
grace_days	INTEGER DEFAULT 7
created_at	TIMESTAMPTZ
updated_at	TIMESTAMPTZ

subscriptions

Column	Type / Constraint
id	UUID, PK
school_id	UUID FK → schools
plan_id	UUID FK → subscription_plans
status	ENUM(trial, active, grace, suspended, archived, cancelled, premier)
started_at	TIMESTAMPTZ
expires_at	TIMESTAMPTZ
cancelled_at	TIMESTAMPTZ NULL
payment_method	ENUM(visa, mtn, orange, bank, manual)
auto_renew	BOOLEAN DEFAULT true
discount_id	UUID FK → discounts NULL
created_at	TIMESTAMPTZ

payments

Column	Type / Constraint
id	UUID, PK
school_id	UUID FK → schools
subscription_id	UUID FK → subscriptions
amount_usd	DECIMAL(10,2)
amount_lrd	DECIMAL(12,2)
exchange_rate	DECIMAL(10,4)
currency_charged	ENUM(USD, LRD)
payment_method	ENUM(visa, mtn, orange, bank, manual)
gateway_ref	VARCHAR(255)
status	ENUM(success, failed, pending, refunded)
recorded_by	UUID FK → admin_users NULL
created_at	TIMESTAMPTZ

invoices

Column	Type / Constraint
id	UUID, PK
school_id	UUID FK
subscription_id	UUID FK
invoice_number	VARCHAR(20) unique
amount_usd	DECIMAL(10,2)
amount_lrd	DECIMAL(12,2)
status	ENUM(draft, sent, paid, overdue, void)
due_date	DATE
paid_at	TIMESTAMPTZ NULL
pdf_url	TEXT NULL
created_at	TIMESTAMPTZ

discounts

Column	Type / Constraint
id	UUID, PK
name	VARCHAR(100)

type	ENUM(percentage, fixed, coupon, school_specific, bulk, seasonal, referral, loyalty)
value	DECIMAL(10,2)
coupon_code	VARCHAR(50) NULL unique
start_date	DATE
end_date	DATE NULL
applicable_plans	UUID[] NULL
max_uses	INTEGER NULL
current_uses	INTEGER DEFAULT 0
stackable	BOOLEAN DEFAULT false
is_active	BOOLEAN
created_at	TIMESTAMPTZ

notifications_log

Column	Type / Constraint
id	UUID, PK
school_id	UUID FK NULL
type	ENUM(trial_warning, payment_due, payment_failed, ...)
channel	ENUM(email, sms, both)
subject	VARCHAR(255)
body	TEXT
status	ENUM(sent, delivered, failed, read)
sent_at	TIMESTAMPTZ

exchange_rates

Column	Type / Constraint
id	UUID, PK
from_currency	VARCHAR(3)
to_currency	VARCHAR(3)
rate	DECIMAL(10,4)
source	ENUM(api, manual)
fetches_at	TIMESTAMPTZ

admin_users

Column	Type / Constraint
id	UUID, PK
email	VARCHAR(255)
name	VARCHAR(100)
role	ENUM(super_admin, billing, support, technical, sales)
is_active	BOOLEAN
last_login	TIMESTAMPTZ NULL
created_at	TIMESTAMPTZ

12. API Endpoints

Method	Endpoint	Description
GET	/api/admin/dashboard	Overview stats
GET	/api/admin/schools	List all schools with filters
GET	/api/admin/schools/:id	School detail with subscription
PATCH	/api/admin/schools/:id/status	Activate/suspend/archive school
GET	/api/admin/plans	List all plans
POST	/api/admin/plans	Create new plan
PUT	/api/admin/plans/:id	Update plan
DELETE	/api/admin/plans/:id	Soft-delete plan
GET	/api/admin/subscriptions	List all subscriptions
POST	/api/admin/subscriptions/:id/extend	Extend subscription
POST	/api/admin/subscriptions/:id/override	Override status
GET	/api/admin/payments	List all payments
POST	/api/admin/payments/record	Record manual payment
POST	/api/admin/payments/:id/refund	Process refund
GET	/api/admin/invoices	List all invoices
GET	/api/admin/discounts	List all discounts
POST	/api/admin/discounts	Create discount/promo
PUT	/api/admin/discounts/:id	Update discount
GET	/api/admin/exchange-rate	Current rate
PUT	/api/admin/exchange-rate	Manual rate override
GET	/api/admin/notifications	Notification log
POST	/api/admin/notifications/broadcast	Send broadcast
PUT	/api/admin/notifications/templates/:id	Edit template
POST	/api/payments/checkout	School-facing: initiate payment
POST	/api/payments/webhook/stripe	Stripe webhook
POST	/api/payments/webhook/mtn	MTN callback
POST	/api/payments/webhook/orange	Orange callback

13. Security & Compliance

13.1 Payment Security

- PCI-DSS compliance via Stripe (card data never stored on our servers)
- Mobile Money tokens stored encrypted, never raw phone numbers in logs
- All payment APIs over HTTPS with TLS 1.3
- Webhook signature verification for all payment callbacks

13.2 Admin Security

- Super Admin login with email + password + 2FA (TOTP)
- Session timeout after 30 minutes of inactivity
- IP allowlisting option for admin dashboard
- All admin actions logged with timestamp, IP, and user ID
- Sensitive actions (delete, suspend) require confirmation

13.3 Data Protection

- Multi-tenant isolation: school_id enforced on every query via RLS
- Encrypted backups daily with 30-day retention
- GDPR-style data export: school can request all their data
- Data deletion: when school is archived beyond retention, data is permanently purged

14. Reporting & Analytics

The Super Admin dashboard includes comprehensive reporting:

14.1 Financial Reports

- Monthly Recurring Revenue (MRR) with trend chart
- Annual Recurring Revenue (ARR) projection
- Revenue breakdown by plan, payment method, currency
- Outstanding invoices aging report
- Refund and churn analysis

14.2 Subscription Reports

- Active subscriptions by plan type (pie chart)
- Trial conversion rate (what % of trials become paid)
- Churn rate (monthly cancellations / total subscribers)
- Subscription growth over time (line chart)
- Schools by status breakdown

14.3 Notification Reports

- Delivery success rates by channel (email vs SMS)
- Open/read rates for email notifications
- Notification-to-payment conversion (did the reminder work?)

14.4 Export Options

- CSV export for all report data
- PDF summary reports (branded with platform logo)
- Scheduled reports: auto-email weekly/monthly summary to Super Admin
- Date range filtering on all reports